

Conversation-Based Guest Check-In via Channel Messaging

Product Specification

v1.0

1. Overview

A fully agent-based private check-in assistant that integrates into any booking application via a REST API endpoint. Guests complete onboarding through a natural conversation with an AI agent. A state machine orchestrates the workflow; the LLM agent handles intent detection, entity extraction, and tool calls — but never advances state directly.

2. Problem Statement

Property managers and booking platforms require guests to complete a multi-step onboarding process before arrival. Current approaches suffer from low completion rates, no real-time assistance, no audit trail, and platform lock-in.

3. Solution

A channel-agnostic, API-first guest check-in system with five core components:

- **State Machine** — workflow orchestrator and single source of truth
- **LLM Agent** — Ollama + Llama 3.1 8B for intent detection and conversation
- **MCP Tools** — 12 internal Python functions for data operations
- **REST API + WebSocket** — API-first integration layer
- **Reference Web Chat Widget** — embeddable JavaScript widget

4. Onboarding Workflow — State Machine

States: INIT → PRIVACY_POLICY_PENDING → HOUSE_RULES_PENDING → RENTAL_AGREEMENT_PENDING → INFO_VERIFY_PENDING → ID_VERIFY_PENDING → INCIDENTAL_PROTECTION_PENDING → COMPLETED

State	Required Action	Valid Response	On Decline
PRIVACY_POLICY_PENDING	Accept/decline		

	Privacy Policy	Agree / Accept / Decline	Record refusal, halt
HOUSE_RULES_PENDING	Accept/decline House Rules	Agree / Accept / Decline	Record refusal, halt
RENTAL_AGREEMENT_PENDING	Accept Rental Agreement	Agree / Accept / Decline	Record refusal, halt
INFO_VERIFY_PENDING	Confirm personal info	Correct / Corrections	Guest provides corrections
ID_VERIFY_PENDING	Upload government ID	Upload via secure link	Must upload to advance
INCIDENTAL_PROTECTION_PENDING	Select Damage Waiver or Security Hold	Selection via link	Must select to advance
COMPLETED	None	N/A	N/A

4.1 State Machine Rules

- The state machine is the single source of truth for guest onboarding progress.
- The LLM agent never advances the state directly — it detects intent and calls MCP tools; the state machine validates and transitions.
- Guests can resume from the last completed step if they pause or disconnect.
- Guests who decline an agreement are recorded as REFUSED and can restart from the beginning.
- Every state transition is logged to the audit trail.

5. LLM Agent Design

Architecture: Guest Message → Intent Classifier → Entity Extractor → Tool Router → State Machine → Response Generator → Guest

Model: Llama 3.1 8B Instruct via Ollama | **Temperature:** 0.3 | **Max tokens:** 1024 | **Context:** 8192 (sliding window)

Capabilities: Intent Detection (agree/decline/question/provide_info/select_option/request_help/greeting/other), Entity Extraction, Tool Invocation, Question Answering (static FAQ), Contextual Prompting.

6. MCP Tools

Tool	Description	State Trigger
get_reservation	Retrieve reservation details	Any
record_agreement	Record agreement acceptance/refusal	Agreement states
update_guest_info	Update guest personal information	INFO_VERIFY
trigger_otp	Send email OTP	INFO_VERIFY
verify_otp	Verify OTP code	INFO_VERIFY
generate_id_upload_link	Generate secure ID upload link	ID_VERIFY
record_id_upload	Record ID upload completion	ID_VERIFY
generate_incidental_link	Generate incidental selection + payment link	INCIDENTAL
record_incidental_selection	Record protection choice + payment	INCIDENTAL
get_arrival_instructions	Generate arrival instructions	COMPLETED
get_faq_answer	Retrieve FAQ answer	Any
get_current_state	Get guest's current state	Any

7. REST API

10 REST endpoints + 1 WebSocket endpoint. API key auth for platform endpoints; session token auth for guest-facing endpoints.

Method	Path	Description
POST	/api/v1/sessions	Create check-in session

GET	/api/v1/sessions/{id}	Get session status
POST	/api/v1/sessions/{id}/messages	Send guest message
GET	/api/v1/sessions/{id}/messages	Get message history
GET	/api/v1/sessions/{id}/state	Get current state
POST	/api/v1/sessions/{id}/resume	Resume paused session
GET	/api/v1/reservations/{ref}	Get reservation details
POST	/api/v1/id-upload	Upload ID document
POST	/api/v1/incidental/select	Select incidental + payment
POST	/api/v1/otp/verify	Verify OTP code
WS	/api/v1/ws/{id}	WebSocket real-time chat

8. Database Schema

MySQL 8.0 with 10 tables: reservations, guests, sessions, agreements, otp_verifications, incidental_selections, messages, audit_trail, knowledge_base, api_keys.

9. Technical Stack

Component	Technology	License
Web Framework	FastAPI 0.110+	MIT
ASGI Server	Uvicorn	BSD
ORM	SQLAlchemy 2.0+	MIT
Migrations	Alembic	MIT
Database	MySQL 8.0	GPL
LLM Runtime	Ollama + Llama 3.1 8B	MIT / Llama 3.1
Email (Dev)	MailHog	MIT
Template Engine	Jinja2	BSD
Validation	Pydantic v2	MIT
Testing	pytest + Playwright	MIT / Apache 2.0
Containers	Docker + Docker Compose	Apache 2.0

10. Security

- API key authentication for platform endpoints
- Session tokens (24h expiry) for guest-facing endpoints
- Secure, one-time-use upload links (1h expiry)
- OTP expiry: 10 minutes, max 3 attempts
- SQLAlchemy ORM (parameterized queries)
- Pydantic input validation
- CORS configuration for chat widget
- Rate limiting: 60 requests/minute per API key

11. Testing Requirements

11.1 Unit Tests

State machine (transitions, decline, resume), Agent (intent detection, entity extraction), MCP tools (each tool, missing records, input validation), API endpoints (validation, auth, schemas).

11.2 E2E Browser Tests

Full happy path, decline path, resume path, question path, ID upload flow, incidental selection flow.

11.3 Integration Tests

WebSocket connect/send/receive, OTP trigger/verify/expiry, API key auth (valid/invalid/missing), DB persistence across restart.

11.4 Test Infrastructure

pytest + pytest-asyncio, Playwright (Chromium), MySQL test DB with seed data, Mock LLM client for non-Ollama tests.

12. Non-Functional Requirements

- Agent response time: <5 seconds (with local Ollama)
- Concurrent sessions: 50 simultaneous on single server
- Availability: 99.5% uptime (single instance)
- Data retention: 12 months audit trail

- Scalability: Stateless API layer for horizontal scaling